

Lesson 27: The Diffusion Equation - Green's Functions

Professor Dave Explains Differential Equations Master Series · Lesson 27

Video: The Diffusion Equation Part 3: Green's Functions **Course:** Applied Ordinary Differential Equations (ENGR 213)

1. Video Intuition & Pedagogical Framing

Professor Dave concludes the Differential Equations master series with the pinnacle of linear operator theory: Green's Functions. Imagine dropping a single concentrated microscopic speck of dye into an infinite pipe at $x = 0$ ($u(x, 0) = \delta(x)$). The dye spreads out into a broadening bell-shaped Gaussian distribution called the Fundamental Solution or Heat Kernel. Because the diffusion equation is linear, ANY arbitrary initial temperature profile $f(x)$ is simply a collection of points; the solution is the continuous superposition (convolution) of heat kernels!

2. Core Theoretical Framework & Rigorous Formulations

1. The Fundamental Solution (Heat Kernel)

The response of $u_t = Du_{xx}$ on $(-\infty, \infty)$ to a point impulse $u(x, 0) = \delta(x)$ is the **Heat Kernel** $\Phi(x, t)$:

$$\Phi(x, t) = \frac{1}{\sqrt{4\pi Dt}} \exp\left(-\frac{x^2}{4Dt}\right) \quad (t > 0)$$

Key properties: (1) Positivity $\Phi > 0$; (2) Total energy conservation $\int_{-\infty}^{\infty} \Phi(x, t) dx = 1$; (3) As $t \rightarrow 0^+$, $\Phi(x, t) \rightarrow \delta(x)$.

2. Green's Integral Representation

For an arbitrary initial Cauchy temperature distribution $u(x, 0) = f(x)$:

$$u(x, t) = (\Phi * f)(x) = \int_{-\infty}^{\infty} \Phi(x - y, t) f(y) dy = \frac{1}{\sqrt{4\pi Dt}} \int_{-\infty}^{\infty} \exp\left(-\frac{(x - y)^2}{4Dt}\right) f(y) dy$$

3. Method of Images for Semi-Infinite Domain

- Dirichlet Boundary ($u(0, t) = 0$): Odd reflection $G(x, y, t) = \Phi(x - y, t) - \Phi(x + y, t)$.
- Neumann Boundary ($u_x(0, t) = 0$, insulated): Even reflection $G(x, y, t) = \Phi(x - y, t) + \Phi(x + y, t)$.

3. Fully Solved Whiteboard Problems

Grand Synthesis Problem: Diffusive Dispersion of an Initial Gaussian Packet

Problem Statement: If initial temperature is already an unnormalized Gaussian $f(x) = e^{-x^2/(4\sigma^2)}$, compute the temperature profile $u(x, t)$ at all subsequent times $t > 0$.

Step 1: Set Up Green's Convolution Integral

$$u(x, t) = \frac{1}{\sqrt{4\pi Dt}} \int_{-\infty}^{\infty} \exp\left(-\frac{(x-y)^2}{4Dt}\right) \exp\left(-\frac{y^2}{4\sigma^2}\right) dy.$$

Step 2: Combine Exponents in y

$$\text{Exponent: } -\left[\frac{(x-y)^2}{4Dt} + \frac{y^2}{4\sigma^2}\right] = -\frac{1}{4} \left[y^2 \left(\frac{1}{Dt} + \frac{1}{\sigma^2} \right) - 2y \frac{x}{Dt} + \frac{x^2}{Dt} \right].$$

Step 3: Complete the Square in y

Let $\frac{1}{S^2} = \frac{1}{Dt} + \frac{1}{\sigma^2} = \frac{\sigma^2 + Dt}{Dt\sigma^2} \implies S^2 = \frac{Dt\sigma^2}{\sigma^2 + Dt}$. Completing the square reveals: $-\frac{x^2}{4(\sigma^2 + Dt)}$ factors cleanly out of the integral!

Step 4: Evaluate Gaussian Kernel Integral

$$\int_{-\infty}^{\infty} \exp\left(-\frac{(y-y_0)^2}{4S^2}\right) dy = \sqrt{4\pi S^2} = \sqrt{\frac{4\pi Dt\sigma^2}{\sigma^2 + Dt}}.$$

Step 5: Multiply Pre-Factors and Cancel

$$u(x, t) = \frac{1}{\sqrt{4\pi Dt}} \sqrt{\frac{4\pi Dt\sigma^2}{\sigma^2 + Dt}} \exp\left(-\frac{x^2}{4(\sigma^2 + Dt)}\right) = \frac{\sigma}{\sqrt{\sigma^2 + Dt}} \exp\left(-\frac{x^2}{4(\sigma^2 + Dt)}\right).$$

Step 6: Physical Interpretation

A Gaussian packet remains a Gaussian packet for all time! Its variance increases linearly: $\sigma^2(t) = \sigma^2 + Dt$ (broadening), while its peak amplitude drops as $\sigma/\sqrt{\sigma^2 + Dt}$ to conserve total thermal energy. The pinnacle of diffusion theory!

⚠ Common Exam Pitfall & Trapdoor Avoidance

Notice how the variance increases as $\sigma^2 + Dt$, NOT as $\sigma + Dt$. Diffusion variance scales linearly with time t , meaning spatial spread scales with $\sqrt{t}$.

4. Key Takeaways & Exam Strategy Formula Card**🎯 Core Takeaway & Exam Rule**

Green's function is the impulse response $\Phi(x, t)$. Any general diffusion solution is the convolution of initial conditions with this fundamental heat kernel.

- Video reference: [The Diffusion Equation Part 3: Green's Functions](#)
- Always verify boundary conditions and check that candidate solutions satisfy the governing differential equation identically.